MILITARY SPECIFICATION

INDICATOR, POSITION ALU-4/A, HORIZONTAL STABILIZER

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of 28V dc integrally lighted horizontal stabilizer position indicator, designated ALU-4/A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Method Of
MIL-C-675	Coating of Glass Optical Elements (Anti-Reflection)
MIL-I-7062	Indicators, Position, Control Surfaces, 28 Volt DC, General Specification For
MIL-T-26609	Transmitter, Position
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification For
MIL-B-27497	Bearing, Jewel, Sapphire or Ruby, Synthetic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6610
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:39Z
Check the source to verify that this is the current version before use.

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-471	Maintainability Demonstration
MIL-STD-794	Part And Equipment, Procedures For Packaging And Packing Of
MIL-STD-810	Environmental Test Methods & Engineering Guidelines
MS24550	Transmitter Position
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form Of
MS33585	Pointers, Dial, Standard Design Of Aircraft Instrument
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to ASTM, 1916 Race St, Philadelphia PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:39Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first articles inspection (see 4.4 and 6.4).

3.2 General specification. The requirements of MIL-I-7062 apply as requirements of this specification with the exception and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 Design. The indictor shall be designed for operation with position transmitters conforming to MIL-T-26609. The pointer shall travel approximately 180 mechanical degrees for 90° rotation of an MS24550-1 transmitter.

3.3.1 Reliability. The indicator shall have a mean-time-between-failures (MTBF) of 225 hours at a 90 percent confidence level.

3.3.2 Maintainability. The following maintainability requirements shall be considered in the design of the indicator:

a. Minimization of complexity of maintenance tasks (for example: calibration, adjustments, inspections et cetera) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items.

b. Required minimum number and types of tools and test equipment (special and standard) needed to perform maintenance.

c. Require minimum number and types of spare parts and assemblies.

3.4 Performance

3.4.1 Fogging. There shall be no evidence of moisture or oil fogging of the cover glass when the indicator is subjected to a temperature of 85° ±5°C for a minimum of 2 hours and then the face of the glass is immediately immersed in an ice bath for 3 minutes.

3.5 Resistance. The dc resistances of the indicator measured between connector pins A, B, and C shall be 475 ±100 ohms. Deviation from the arithmetic mean of the resistance values between the coils shall not exceed 1.500 percent of the arithmetic mean.

3.6 Case. The case shall have a 1.500 inch nominal diameter and shall conform to figure 1 and MS33639. The case shall be hermetically sealed.

3.6.1 Filling medium. The filling medium shall be either 100 percent helium or a mixture of 88 to 92 percent nitrogen and the remainder helium. Where practicable 100 percent helium shall be used at a pressure of approximately 1 atmosphere.

3.7 Dial. The dial shall conform to figure 1 and shall be securely fastened to the frame of the mechanism in such manner that it will not lossen nor turn when the indicator is vibrated. The diameter of the dial shall be not less than 1.313 inches as measured across the ends of the graduations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:39Z
Check the source to verify that this is the current version before use.

The style and proportions of numerals and lettering shall conform to MS33558. Each numeral shall distinctly indicate to which graduation it applies.

3.7.1 Marking. The dial markings shall be durable and in accordance with Table I. The colors shall be in accordance with FED-STD-595.

TABLE I. DIAL MARKINGS.

Marking Marking	Height or Length ±0.010 (Inch)	Width of Line or Graduation ±0.005 (Inch)	Color No.
Lettering	0.094	0.020	37875
HORIZ STAB DEGREES	0.125	0.020	37875
NOSE UP, NOSE DN	0.094	0.020	37875
Numerals 0, 4, 8, 12	0.156	0.031	37875
Major graduations	0.125	0.020	37875
Minor graduations	----	----	37875
Shaded portion of pointer			
Lettering ALU-4/A	0.062	----	37038
Background of dial	----	----	37038

3.7.2 Antireflection coating. All reflecting glass surfaces shall be provided with a reflection reducing coating which meets the requirements of MIL-C-675, in addition to withstanding the environmental conditions specified herein, except that the following reflectance tolerances shall apply:

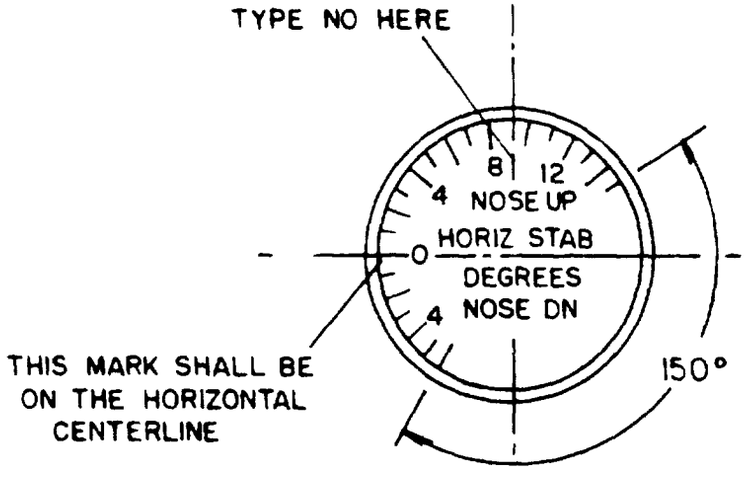
Angle of Incidence	Wavelength Millimicrons	Average Percent Reflectance
0°	450 to 675	0.6
0°	425 to 700	0.5
30°	450 to 625	1.0
30°	425 to 700	0.5

3.8 Pointer. The pointer shall be in accordance with MS33585-8 and suitable for integral lighting.

3.9 Electrical connector. A Bendix PTIH-10-6P, or equivalent, electrical connector shall be provided.

3.10 Lighting. The indicator shall be integrally lighted in accordance with MIL-I-27160. The lighting system shall be so designed that replacement lamps may be accurately positioned.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:39Z
Check the source to verify that this is the current version before use.



DIAL INDICATIONS		DEGREES FROM HORIZ ℄	
0		0	
1	UP	10	CW
2		20	
3		30	
4		40	
5		50	
6		60	
7		70	
8		80	
9		90	
10		100	
11		110	
12		120	
13		130	
14		140	
15	UP	150	CW
1	DN	10	CCW
2		20	
3		30	
4		40	
5		50	
6	DN	60	CCW

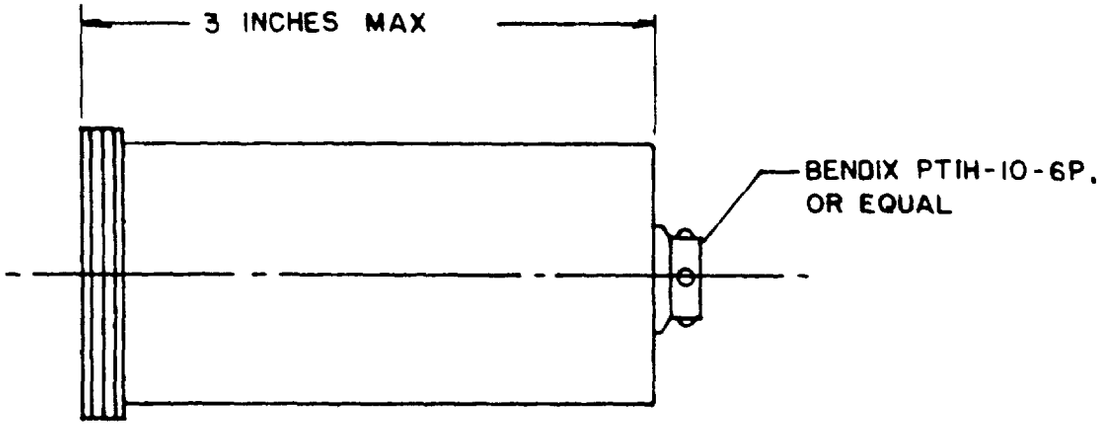


FIGURE 1 Case.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:39Z
Check the source to verify that this is the current version before use.

3.11 Jewel bearings. When synthetic jewel bearings are used, they shall conform to MIL-B-27497.

3.12 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection will be as specified in MIL-I-7062, except that where qualification is specified, first article shall apply.

b. Quality conformance inspection method of sampling and inspection shall be as specified in MIL-I-7062.

4.2.1 General specification. The inspection and testing of indicators shall be in accordance with MIL-I-7062 except as specified herein. For the purpose of this specification, the tests shall be identified as follows:

MIL-I-27672
First article inspection
Individual tests
Sampling plan A
Sampling plan B

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:39Z
Check the source to verify that this is the current version before use.

4.3 Inspection condition. Unless otherwise specified all inspections shall be performed in accordance with the test conditions specified in 4.3.1 this specification.

4.3.1 Scale and friction error tolerances. The scale and friction error shall be read from the test transmitter scale with the indicator pointer on each major graduation of the indicator dial. The errors shall not exceed those specified in Table II.

TABLE II. SCALE AND FRICTION ERROR TOLERANCES.

	Scale Error Mechanical (Degrees)	Friction Error Mechanical (Degrees)
Room temperature	±4	±3
Position	±2	
Low temperature	±5	±4
High temperature	±4	±4

4.4 First article inspection

4.4.1 First article samples. The first article samples shall consist of three indicators. Samples shall be identified with the manufacturer's own part number and any additional information required by the letter of authorization.

4.4.2 First article inspection. The first article inspection shall consist of all the tests specified herein. For the purpose of this specification, First article tests shall be identified in MIL-I-27672 and MIL-I-7062.

4.5 Test methods

4.5.1 Individual tests.

4.5.1.1 Lighting. The indicator shall be subjected to the individual tests specified in MIL-L-27160.

4.5.1.2 Sealing. The instrument case shall be tested for leaks by means of a mass-spectrometer type of helium leak detector. Where a nitrogen-helium mixture is utilized as a filling medium, the detected leak rate shall not exceed 0.010 micron cubic foot per hour at a pressure differential of approximately 1 atmosphere. Where 100 percent helium is utilized as the filling medium, the leak rate shall not exceed 0.10 micron cubic foot per hour.

4.5.2 Sampling plan A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:39Z
Check the source to verify that this is the current version before use.

4.5.2.1 Indicator. The indicator shall be subjected to the sampling plan A tests specified in MIL-L-27160.

4.5.2.2 Vibration error. The indicator shall be mounted on the vibration stand in the normal operating position. It shall be energized with 28V dc and the lighting system shall be energized with 5V dc. The indicator shall then be subjected to a vibration test in accordance with MIL-STD-810, Method 514.2, category b.1, curve B, except that from 300 to 500 Hz, the g-level shall be 1.0g in lieu of 2.0g. Below 300 Hz, the g-level shall be 2.0g. Before vibration is started, the position transmitter shall be adjusted so that the indicator pointer is at the 0° position and shall not be readjusted after vibration has been started. During vibration, the maximum pointer oscillation shall not exceed 10 mechanical degrees double amplitude and the pointer shift (measured from the center of pointer oscillation) shall not exceed 4 mechanical degrees from the original 0° position setting. All lamps shall be operating at the completion of this test. The indicator shall then be subjected to the scale error test. No looseness in the mechanism or damage to any part of the indicator shall result from this test.

4.5.3 Sampling plan B. The indicator shall be subjected to the sampling plan B tests specified in MIL-L-27160.

4.5.3.1 Vibration failure and cycling. The three samples indicators shall be subjected to a combination vibration and pointer cycling tests for a period of 115 hours. A vibration machine shall be used that will provide a double amplitude of 0.005 to 0.02 inch at a frequency of 45 to 60 Hz. The amplitude and frequency chosen shall provide a vibration level of 1.5g $\pm$10 percent. The direction of vibration shall be perpendicular to the longitudinal axis of the indicator. Before starting this test, the indicators shall be subjected to the scale error and friction error tests. Pointer cycling shall consist of pointer deflection from the 6° below 0 position to the 12° position above 0 and return to the 6° below 0 position. This deflection shall constitute 1 cycle and shall be repeated for a period of 115 hours at a rate of not less than 3 Hz or more than 4 Hz. The vibration specified shall be applied for 15 minutes during each hour. Upon completion of this test, the indicators shall again be subjected to the scale and friction error tests. The change in scale and friction error at the various test positions shall not exceed 2 mechanical degrees. There shall be no looseness in the mechanism, damage to, or malfunction of the indicators during or as a result of this test.

4.5.3.2 Reliability. Satisfactory completion of all the tests specified herein, with no failure, will demonstrate an MTBF of 225 hours at a 90 percent confidence level.

4.5.3.3 Fogging. The indicator shall be placed in a controlled ambient temperature of $85^{\circ} \pm 5^{\circ}$C for a minimum of 2 hours. Within 5 seconds after removal from the oven, the indicator face shall be immersed to a depth of 0.250 to 0.333 inch in an ice bath maintained at 0° to 5°C. After 3 minutes, the indicator shall be removed from the ice bath and quickly wiped dry (compressed air shall not be used. The cover glass and dial shall immediately be inspected for evidence of fogging or damage. Evidence of fogging or damage shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:39Z
Check the source to verify that this is the current version before use.

4.5.4 Reflectance. The reflectance of all reflecting glass surfaces shall be measured and shall meet the reflectance tolerances specified in 3.7.2.

4.5.5 Maintainability verification. Maintainability verification shall be in accordance with MIL-STD-471.

4.5.6 Inspection of preservation, packaging, packing, and marking. The preservation, packaging, packing, and marking of indicators shall be inspected to determine compliance with section 5 herein.

4.6 Inspection of packaging. The inspection of the preservation-packaging and interior package marking shall be in accordance with group A and B quality conformance inspection requirements, section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 or MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Each indicator shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density and thickness to insure shock transmission does not exceed peak values in G's established for the indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of indicator shall be in accordance with ASTM D3951.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:39Z
Check the source to verify that this is the current version before use.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistant with the protection required.

5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The ALU-4/A position indicator covered by this specification is intended for use in indicating the position of the horizontal stabilizer in degrees of travel from the neutral position and for use where an integrally lighted indicator is required. The position transmitters used with this indicator are connected to the stabilizer actuating mechanism so that the transmitter shaft is rotated 5 mechanical degrees for 1° stabilizer travel.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. When sampling plan B tests will not be conducted.

c. Levels of packaging and packing required.

6.3 Definitions. Definitions will be in accordance with MIL-I-7062 except that the term double amplitude will mean the extent of motion as measured from one extreme to the opposite extreme.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:39Z
Check the source to verify that this is the current version before use.

6.4 First article. A first article is required for inspection and it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Aquisition Regulation. First article tests are described in 4.4. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

6.5 Subject term (key word) listing.
ALU-4/A
Horzontal Stablizer
Indicator
Position

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
AIR FORCE -99

Review activities
AIR FORCE -11

Preparing activity:
AIR FORCE -71

Project number:
6610-F164

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:39Z
Check the source to verify that this is the current version before use.

submitting comments and suggestions for improvements All users of military standardization documents are invited to provide suggestions This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered

NOTE This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

☆ U S Government Printing Office 1982—360-979/5804

(Fold along this line)

(Fold along this line)

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 73236 WASH DC

POSTAGE WILL BE PAID BY HQ U S AIR FORCE

OC-ALC/MMEDO
Tinker AFB OK 73145-5994

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:39Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-I-27672B	

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR
	☐ USER
b ADDRESS (Street, City, State, ZIP Code)	☐ MANUFACTURER
	☐ OTHER (Specify) _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER (Last, First, MI) - Optional	b WORK TELEPHONE NUMBER (Include Area Code) - Optional
c MAILING ADDRESS (Street, City, State, ZIP Code) - Optional	8 DATE OF SUBMISSION (YYMMDD)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T12:39Z
Check the source to verify that this is the current version before use.